

Virtual Reality One-Session Treatment of Child-Specific Phobia of Dogs: A Controlled, Multiple Baseline Case Series

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Specific phobia (SP) typically onsets in childhood and frequently predicts other mental health disorders later in life. Fortunately, childhood SP can be effectively treated with cognitive behavior therapy (CBT), including the exposure-based one-session treatment (OST) approach. Despite empirical support for CBT and OST, clinicians, for various reasons, frequently fail to implement exposure-based therapy in routine clinical practice, including perceived difficulties in implementing exposure. Virtual reality (VR) exposure therapy may overcome some of these challenges and provides an alternative modality of therapy. This preliminary study examined the efficacy of VR OST for 8 children with a SP of dogs (aged 8–12 years) ($M = \text{age } 10.25$; $SD = 2.11$) using a multiple-baseline controlled case series. Following a stable baseline period of either 2, 3, or 4 weeks, it was expected that specific phobia severity would significantly decline after VR OST and remain improved over the 3-week maintenance phase. Assessments were conducted posttreatment and at 1-month follow up (study end-point). It was found that phobia symptoms remained relatively stable across the baselines, with

significant reductions from pretreatment to posttreatment and to follow-up on clinician severity ratings (pre- to post- $g = 1.12$; pre- to follow-up $g = 2.40$), target symptom ratings ($g = 1.14$; 1.29), and behavioral avoidance ($g = -1.27$; -1.96). The treatment was also associated with clinically significant outcomes, whereby at one-month follow up, 75% of children were considered “recovered” and 88% completed the BAT (interacted with their feared stimuli). This study provides support for the effectiveness of VR OST.

Keywords: specific phobia; one session treatment; virtual reality; children

SPECIFIC PHOBIA (SPs) ARE among the most common mental health disorders affecting children, adolescents, and adults, with estimated lifetime prevalence rates approaching 30% (Egger & Angold, 2006; Merikangas et al., 2010). Aside from the significant distress and impairment associated with this disorder (Ialongo, Edelsohn, Werthamer-Larsson, Crockett, & Kellam, 1995), evidence from prospective longitudinal studies (Bittner et al., 2004; Gregory et al., 2007; Lieb, Miché, Gloster, Beesdo-Baum, & ... & Wittchen, H. U., 2016;) suggests that SPs in childhood may be a gateway disorder and serve as a powerful marker of risk for the development of mental health disorders later in life. Importantly, SPs can be treated in a single

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session of exposure therapy, with robust evidence in support of a one-session treatment (OST; Ollendick et al., 2009, 2015; Öst, Svensson, Hellström, & Lindwall, 2001) for children and adolescents.

OST is a variant of cognitive-behavioral therapy (CBT), and combines graduated in-vivo exposure therapy (EXP), participant modeling, social reinforcement, psychoeducation, and cognitive challenges in one intensive 3-hour-long treatment session, proving an extremely time-limited and efficient therapy (Davis, Ollendick, & Öst, 2012, 2019). Currently designated as a *well-established* treatment (Chambless & Ollendick, 2001), OST has empirical support derived from three randomized controlled trials (RCTs) (Ollendick et al., 2009, 2015; Öst et al., 2001), several smaller clinical trials of youths 7 years and over (e.g., Farrell, Kershaw, and Ollendick, 2018; Farrell, Waters, et al., 2018; Waters et al., 2014), and a controlled case series with young children 4 years of age (Farrell, Kershaw, and Ollendick, 2018; Farrell, Waters, et al., 2018). Across studies, OST has been found to be superior to waitlist (Ollendick et al., 2009; Öst et al., 2001), psychological placebo (Ollendick et al., 2009), and Eye Movement Desensitization and Reprocessing therapy (Muris, Merckelbach, Holdrinet, & Sijsenaar, 1998; Muris, Merckelbach, & Collaris, 1997).

Despite the availability of empirically supported CBT (and OST in particular), there remains a high rate of unmet need in the community, with as few as 2% of youths with mental health problems receiving specialist, evidence-based interventions (Lawrence et al., 2016). Exposure therapy (EXP), the process of gradually exposing a child to feared stimuli without escape or avoidance behaviors, is the essential ingredient in CBT for treating child anxiety and SPs, with recent meta-analyses and dismantling studies (Ale, McCarthy, Rothschild, & Whiteside, 2015; Whiteside et al., 2015; Whiteside, Deacon, Benito, & Stewart, 2016) demonstrating the greater importance of EXP relative to other CBT techniques. Despite this, EXP continues to be one of the least implemented treatment strategies in community settings (Böhm et al., 2008; Cook, Biyanova, Elhai, Schnurr, & Coyne, 2010; Pittig, Kotter, & Hoyer, 2019; Trask, Fawley-King, Garland, & Aarons, 2016), even when community therapists treating child anxiety endorse a CBT orientation (Whiteside et al., 2016). Reasons for this reluctance include inadequate training (Reid et al., 2017; Reid et al., 2018), negative attitudes about EXP (i.e., perceived as potentially damaging or unethical; Olatunji, Deacon, & Abramowitz, 2009), and the perceived stress associated with implementation (Pittig et al., 2019; Reid et al.,

2018). Furthermore, community therapists rate EXP as the most difficult CBT strategy to deliver (Cartreine, Ahern, & Locke, 2010).

Moreover, while OST offers a highly efficient approach to delivering EXP, it is not without its challenges. For example, practical challenges associated with OST implementation include requiring access to various phobic stimuli (e.g., insects; animals; heights; water), which adds a complexity and potential burden on therapists attempting to deliver this evidence-based treatment in routine practice. Furthermore, some phobic stimuli are particularly difficult to access (e.g., flying, storms, water). Beyond implementation challenges, patient preferences can also be a barrier, with patients' negative attitudes about EXP deterring them from accessing therapy (Marks, 1992). Thus, novel approaches aimed at overcoming barriers to implementation of EXP, whilst retaining the therapeutic value of EXP, are needed.

VIRTUAL REALITY EXPOSURE THERAPY

One approach to improving the implementation of EXP in community practice may be via the use of virtual reality (VR) technology. VR is an application that, in very near real time, allows a user to navigate through, and interact with, a virtual environment (Pratt, Zyda, & Kelleher, 1995). A recent review of VR technology in psychological treatments highlights the advantages and effectiveness of VR across various disorders, but particularly for the treatment of anxiety (Maples-Keller, Bunnell, Kim, & Rothbaum, 2017). Importantly, VR delivery of EXP may address some of the implementation barriers, particularly the practical challenges to delivering EXP, as well as providing evidence-based options for patients in how they receive EXP. Indeed, research suggests that patients with SPs report a preference for VR EXP over in-vivo EXP, and moreover, are less likely to refuse VR EXP relative to in-vivo EXP (see Garcia-Palacios, Botella, Hoffman, & Fabregat, 2007). For therapists, VR EXP may present a more manageable, efficient means of delivering EXP without the need to leave the clinic room or manage multiple phobic stimuli (including housing outside of the session), and may facilitate EXP when in-vivo EXP is too challenging due to the accessibility and/or costs of acquiring stimuli (i.e., flying, storms, snakes). In a recent survey of practicing community CBT therapists' attitudes towards VR EXP, therapists ($n = 185$) reported more pros (e.g., exposure to stimuli that could only be delivered in a virtual setting; more precise control over stimuli) than cons (e.g., the virtual setting not being real enough; gains not translating to real-world improvements) in the use

of VR EXP treatment for anxiety disorders (Lindner et al., 2019). While VR EXP does not address all of the known barriers to improving the implementation of EXP, it addresses some, and, moreover, allows for therapist and client preferences in *how* this often confronting, yet effective therapeutic approach is delivered and received.

VR technology has been around for quite some time, but, to date, has not been extensively implemented in clinical practice. This may in part be due to poor quality of earlier VR technology, and difficulty accessing such technology outside of research settings (see Coelho, Waters, Hine, & Wallis, 2009, for a review). However, as VR technology has improved and become more affordable, these modern systems have the potential to target therapeutic mechanisms that cannot be easily achieved with in-vivo EXP. For instance, EXP to stimuli with VR allows therapists to pause, repeat, and restart EXP, which may provide for enhanced extinction learning. Furthermore, given recent findings that EXP to multiple feared stimuli and in different contexts is associated with enhanced retention of extinction learning (Rowe & Craske, 1998; Shiban, Schelhorn, Pauli, & Mühlberger, 2015; Waters, Kershaw, & Lipp, 2018; Weisman & Rodebaugh, 2018), VR EXP may overcome numerous practical challenges to incorporating these evidence-informed procedures into practice.

VIRTUAL REALITY EXPOSURE THERAPY FOR SP

To date, there have been limited RCTs examining VR EXP as a treatment for SPs. Michaliszyn, Marchand, Bouchard, Martel, and Poirier-Bisson (2010) conducted an RCT of VR EXP treatment of spider phobia in 16 French-speaking adults in Montreal. The pilot study compared VR EXP to in-vivo EXP and a waitlist control. Findings demonstrated clinically and statistically significant improvements for the in-vivo EXP condition relative to VR EXP; however, this was only found on one self-report measure of spider fears. Further, the VR EXP condition resulted in statistically and clinically significant improvements in phobia severity relative to the waitlist condition. Since this early study, there have been numerous case studies and small trials that have examined the use of VR EXP in the treatment of SP in adults, providing promising findings. Morina, Ijntema, Meyerbröker, and Emmelkamp (2015) published a meta-analysis of 14 clinical trials ($N = 265$) examining VR EXP for SP in adults and found that in addition to improvements in behavioral assessments immediately following treatment, there were no significant differences between VR EXP and in-vivo EXP conditions. The authors concluded that

VR EXP is an effective modality of implementation for adult SP, with outcomes translating to behavior change in real life settings. Similarly, a more recent meta-analysis of VR EXP for anxiety disorders (Carl et al., 2019) including 30 RCTs ($n = 1,057$ participants), 14 of which involved samples of adults with SPs, found VR EXP was associated with large effect sizes relative to waitlist controls ($g = 0.90$), medium to large effect sizes relative to psychological placebo ($g = 0.78$), and, of most significance, there were no significant differences between VR EXP and in vivo conditions ($g = -0.07$).

Currently, there are comparatively fewer studies with children, with limited evidence for the efficacy and/or feasibility. In one study aimed at treating child SP, Bouchard, St-Jacques, Robillard, and Renaud (2007) conducted a pilot research trial over 14 weeks of the effectiveness of VR EXP for child SP of spiders ($n = 9$, mean age 11.3 years). Following four sessions of VR EXP, children self-reported a significant reduction in spider fears. In a later RCT, St-Jacques, Bouchard, and Bélanger (2010) examined VR treatment for SP of spiders among 31 children (8–15 years of age). Participants were randomized to receive either an in-vivo EXP treatment, or a four-session VR EXP plus one session in-vivo EXP condition. In addition to assessing the relative treatment outcomes of in-vivo versus VR EXP, they also aimed to determine whether the use of VR would increase the child's motivation towards therapy. Assessments of extrinsic motivation and reluctance to attend therapy revealed no significant increase in motivation across either condition. However, similar to previous studies, results showed that both in-vivo and VR EXP equally resulted in significant reductions in fear. In the only other RCT of VR EXP for child SP, Maskey et al. (2019) conducted an RCT of one-session CBT followed by 4 VR EXP sessions relative to a care-as-usual (CAU) among 32 children (8–14 years) with a primary diagnosis of Autism Spectrum Disorder, and comorbid SP. At 6 months postassessment, 38% of children in the VR EXP condition were considered responders, relative to none in the waitlist; furthermore, five children in the CAU experienced deterioration in their SP relative to only one child in the VR EXP condition. Collectively, these preliminary findings demonstrate that VR EXP may be feasible and efficacious for both neurotypical children, as well as youth with ASD. Currently however, no studies have examined the efficacy of a VR OST for childhood SP.

THE CURRENT STUDY

While in vivo OST is supported by empirical evidence, few patients receive this treatment due

Table 1
Participant Demographics

Participant	Age	Gender	Ethnicity	Primary Diagnosis	Secondary Diagnosis	Tertiary Diagnoses	Number of diagnosis
1	8	M	Caucasian	SP Dog	N/A	N/A	1
2	9	M	Caucasian	SP Dog	N/A	N/A	1
3	12	F	Caucasian	SP Dog	SP Cat	SP BII	3
4	12	F	Caucasian	SP Dog	SP Loud Noise	N/A	2
5	10	F	Caucasian	SP Dog	SP Cat	N/A	2
6	11	M	Caucasian	SP BII	SP Dog	Generalized Anxiety Disorder	3
7	9	M	Caucasian	SP Dog	SP Insects, Water, Dark, BII, Toilets	Generalized Anxiety Disorder, Attention Deficit Hyperactivity Disorder, Conduct Disorder	10
8	12	F	Caucasian	SP Dog	SP Water	N/A	2

Note: M = Male; F = Female; SP = Specific Phobia

to barriers in accessing EXP in the community (Ollendick, Ryan, Capriola-Hall, Fraire, & Austin, 2018). Given the cumulative evidence for the comparable efficacy of VR EXP relative to in-vivo EXP among adults, combined with evidence that VR may offer some advantages for targeting therapeutic mechanisms, is more affordable than in the past, and clinicians and patients may prefer VR EXP to in-vivo EXP, studies aimed at testing the feasibility and efficacy of VR OST for child SP are warranted. This study examined whether VR OST results in clinically significant improvement for children with an SP of dogs using a controlled, multiple baseline case series design where participants are randomly assigned to 2-week, 3-week, or 4-week baselines, followed by the VR OST, and a 1-month follow-up. It was hypothesized that specific phobia symptoms (clinician rated severity and parent-rated target symptoms) will remain stable during the baseline and will significantly improve following VR OST, and remain stable (and low) across 1-month follow-up. Furthermore, it was hypothesized that behavioral approach (as measured using a standardized Behavioral Approach Task [BAT]) would significantly improve immediately following VR OST relative to baseline and remain improved across 1-month follow-up. It was hypothesized that symptomatic changes would be clinically significant as indexed by the reliable change criterion (Jacobson & Truax, 1991). Finally, it was hypothesized that there would be significant improvements across time (to post- and 1-month follow-up) on secondary outcomes, including child and parent-rated anxiety, and child-rated fear.

Method

PARTICIPANTS

Participants were 8 children ($n = 4$ males, 4 females), aged 8 to 12 ($M = 10.25$; $SD = 2.11$) years, with a

clinical diagnosis of a specific dog phobia, according to the DSM-5 criteria (American Psychiatric Association, 2013). Inclusion criteria included: a diagnosis of dog phobia, aged 8 to 12 years, one parent willing to attend treatment, and, if on medication, a stable dose for 12 weeks prior (no child was on medication). Exclusion criteria included a nonanxiety primary diagnosis, diagnosis of Autism Spectrum Disorder, Intellectual Impairment, psychotic symptoms or high suicidal ideation. Seventy-five percent ($n = 6$) had comorbid diagnoses (range 1–9 comorbid disorders), with other SPs the most common comorbidity. Table 1 presents demographic and diagnostic information for the sample.

DESIGN

The current study utilized a nonconcurrent, multiple baseline case series design, whereby following a pretreatment assessment, children were randomly assigned to 2-week, 3-week, or 4-week baseline assessment phases representing a series of A-B replications (Hayes, Barlow, & Nelson-Gray, 1999; Kazdin & Weisz, 1998). Single-case designs have been endorsed by the evidence-based treatment movement (Task Force on Promotion and Dissemination, 1995) and allow for experimental control when testing novel interventions (e.g., Farrell et al., 2016; Farrell, Kershaw, and Ollendick, 2018; Farrell, Waters, et al., 2018; Oar, Farrell, Waters, Conlon, & Ollendick, 2015). The multiple baselines control for the effects of time, as well as a comparison of each child's baseline serving as their own within-subjects control. Outcome variables included: diagnosis and clinician severity rating (CSR), parent-rated target symptoms, behavioral approach task (BAT), and self-report scales (SCAS-P/SCAS-C; FSSC-R-C). Baseline variables included: CSR and parent target symptom ratings.

MEASURES

Anxiety Disorders Interview Schedule: Parent version (ADIS-P; Silverman & Albano, 1996)

The ADIS-P assesses anxiety disorders and commonly occurring comorbidity in children (Silverman & Eisen, 1992). The ADIS-P has demonstrated good sensitivity to treatment effects in childhood anxiety and SP (Ollendick et al., 2009). Each diagnosis receives a Clinician Severity Rating (CSR), scored 0–8, with a score of 4 and greater indicating a clinical diagnosis. CSR ratings by our trained assessors have been established as excellent (e.g., $\kappa = .94-.86$ across primary, secondary and tertiary diagnoses; see Farrell, Kershaw, and Ollendick, 2018; Farrell, Waters, et al., 2018).

Target Symptoms

Three individualized target symptoms were obtained for each child, which represented three primary symptoms associated with the child's SP. In the baseline period, target symptoms were rated by parent/s indicating the child's level of fear associated with each symptom (e.g., sight of a dog, seeing a dog run towards the child) on a scale ranging from 0 to 8 (*how fearful?* 0 = none to 8 = very much). The child's most severe target behavior was reported (Farrell et al., 2016; Ollendick et al., 2009).

Behavioral Approach Tests (BATs)

Each child completed a standardized BAT at pretreatment, posttreatment and 1-month follow-up, whereby the child was asked to enter a large room and approach a dog being held on a lead by a dog owner in the far corner, and pat the dog on the head for 20 seconds. Each child received standardized instructions. The dog used for the BAT was the same dog at pretreatment as posttreatment for each child (however, dogs varied from child to child, depending on the availability of dog volunteers). The BAT dogs were different dogs to those used in the VR OST treatment session. Behavioral approach was rated by the experimenter on a Likert-scale of 0 to 10 with a score of 10 indicating the child completed the BAT. A score of 0 denoted the child not opening the door, with each score in between denoting a step closer to the goal. These procedures replicate those of previous studies (e.g., see Oar et al., 2015; Ollendick et al., 2009).

Spence Children's Anxiety Scale Child and Parent Versions (SCAS-C/P; Spence, 1998)

The SCAS-C/P is a self- and parent-report assessment of anxiety symptoms for children 7 to 18 years. The SCAS-C/P has been found to have good to excellent reliability (Cronbach's alpha for total .89 - .92) as well as validity (Nauta et al., 2004; Spence, 1998).

Fear Survey Schedule for Children-Revised Child Version (FSSC-R-C; Ollendick, 1983). The FSSC-R-C is a self-report measure to assess fearfulness in children aged 7 to 16 years. The measure possesses well-established reliability and validity estimates and provides norms for various ages and nationalities (Ollendick, 1983; Ollendick, King, & Frary, 1989).

Perceived Reality of VR Stimuli

During VR OST, children were asked hourly to rate the perceived reality of stimuli. Rating were made on a 5-point Likert scale from 0 (*not at all like real life*) to 4 (*very real*).

STIMULUS MATERIAL AND VR TECHNOLOGY

The VR stimuli was created using a Fly 360 4K camera in the same room the OST was to be conducted at the university psychology clinic. The use of a 360-degree camera and headset allows users the freedom to look all around them, as if in the environment itself, and thus provides a more immersive delivery than regular footage on a video screen. Six different dogs (Doberman, English Cocker Spaniel, Labrador x Kelpie, Rottweiler x Border Collie, Cavoodle, and Japanese Spitz) were used for footage to provide multiple exemplars of the phobic stimulus (cf. Waters et al., 2018). Each dog was selected based on providing a variation of breeds and sizes to maximize variability. Approximately 1 hour of footage was taken for each dog and Adobe premiere, Adobe after effects, and Adobe media encoder programs were used to edit the footage. The Oculus Rift VR headset was used for the participants to view the footage with the Oculus computer program. During VR OST, SkyBox was used so that the therapist could see what the child was seeing.

PROCEDURE

Pretreatment

Following university ethical approval (GU Ref No: 2018/143), prospective participants were recruited through advertisements in local independent school newsletters and through media coverage, as well as social media advertising. Once the child was assessed for suitability of the study via a brief telephone screen, the study information and consent forms were emailed, which were completed at the initial clinic assessment. Parents completed the full ADIS-P interview over the phone to determine eligibility (providing verbal consent prior to the interview). One eligible child declined to participate due to a scheduled family vacation. See Figure 1 for the Consort diagram. At the subsequent clinic assessment session, the parent-rated target symptoms, SCAS (parent and child), FSSC-R-C (child rated), and BAT were completed. All pretreatment assessment measures were completed by a

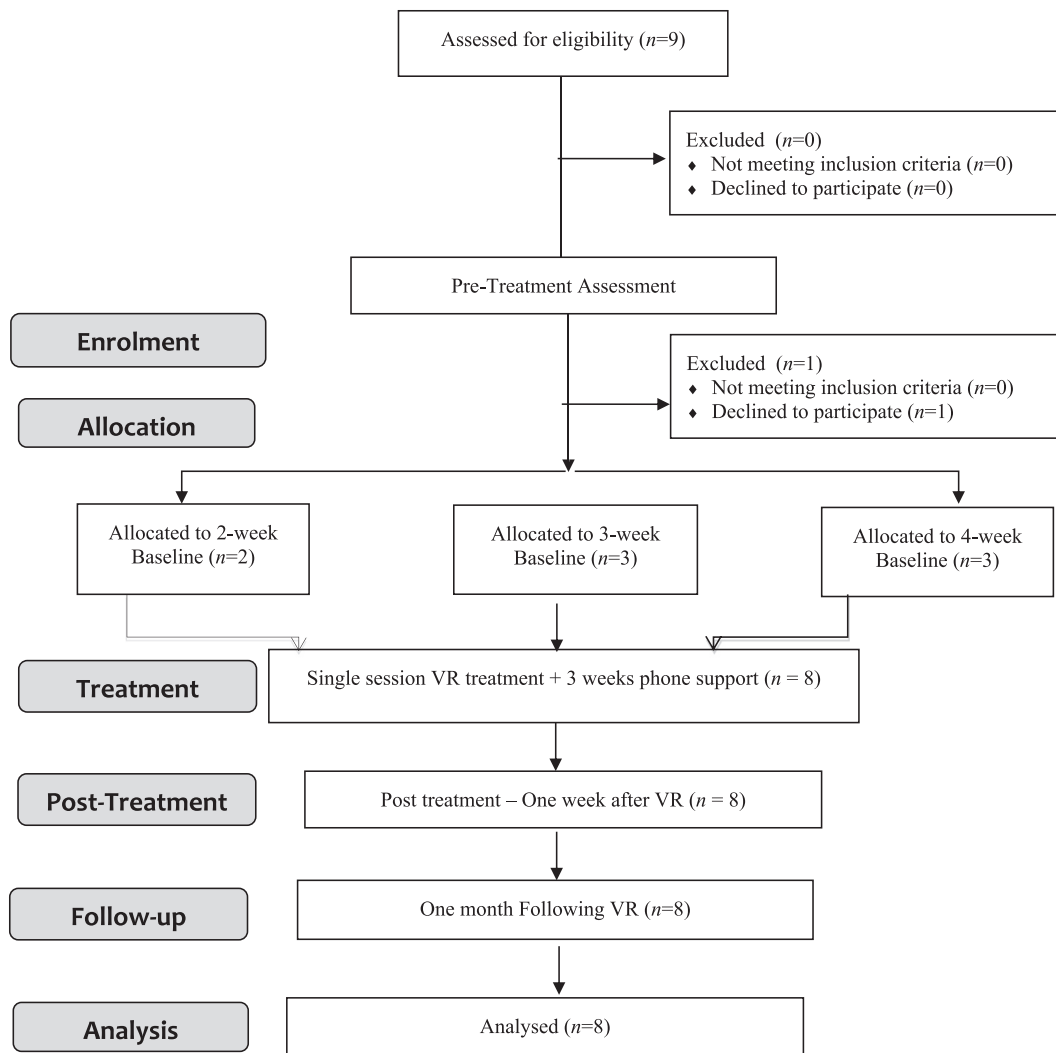


FIGURE I Participant Flow Diagram

trained, postgraduate clinical psychology student who also delivered all of the treatment sessions under the supervision of the first author (LJF). The participants were then allocated randomly into one of three different baseline groups; whereby they completed a 2-week baseline, 3-week baseline, or 4-week baseline.

Baseline

On a weekly basis during the baseline period (2, 3 or 4 weeks), parents completed a telephone-administered ADIS-P (SP Dogs module only) and a CSR rating was obtained. Additionally, parent-rated target symptoms were collected weekly over the baseline.

Virtual Reality One-Session Treatment (VR OST)

The treatment followed OST principles described by Ollendick et al. (2009) and Öst (1997) and was flexibly delivered to each child by a therapist

trained in OST. The treatment session involved 3 hours of graduated VR exposure therapy. At the beginning of the session and prior to equipping the VR headset, the children's phobic beliefs were assessed and they were familiarized with a measure of subjective units of distress (SUDs), which was used throughout the session to aide in pacing exposure tasks. Prior to seeing a dog in the VR footage, the child was permitted to become accustomed to the VR headset. Children were instructed that they were not to remove the headset during the session, unless they started to feel physically unwell, in which case they could take the headset off and take a short break. They were informed that if the stimuli were very distressing, they would instead slow the exposure rather than removing the headset. Children and parents were informed that there would be breaks approximately

Table 2
Hierarchy Steps for Each Dog Within the VR EXP

Steps	Stimuli
1	Dog and assistant walks into and sits on the opposite side of the room (on leash)
2	Subject moves closer to dog (on leash)
3	Subject moves closer to dog (on leash)
4	Subject moves directly next to dog (on leash)
5	Subject back to original side of room, assistant and dog standing up walking 1m forward (on leash)
6	Assistant and dog standing up walking 1m forward from previous position (on leash)
7	Assistant and dog standing up walking 1m forward from previous position (on leash)
8	Dog walking side to side and around camera (on leash)
9	Dog walking/running towards subject (off leash and assistant in room)
10	Dog without assistant in room and no leash

Note: VR EXP = Virtual Reality Exposure Therapy

every 50 minutes, at which time they could take the headset off.

All participants underwent the same sequence of VR EXP steps; however, they progressed through the treatment at different rates. Throughout treatment, the therapist was able to monitor what the child was seeing. Thus, during the treatment, the therapist asked questions regarding the dog, the child's phobic beliefs, and SUD ratings. This allowed the therapist to confirm the child was not performing avoidant behaviors such as closing their eyes or looking away, given that visual avoidance of the feared stimulus even when it remains in the visual field interferes with extinction learning (O'Malley & Waters, 2018). Table 2 shows hierarchy steps for each dog.

The child's SUDs were taken approximately every 5 minutes throughout the session and exposure tasks only progressed to the following step after at least a 50% reduction in their original SUDs, or if a step was deemed too easy/comfortable for the child (e.g., Ollendick et al., 2015; Ollendick et al., 2018). In line with the OST approach, psychoeducation was also provided throughout the session to provide information on dog behavior, and how to interact safely with dogs in real life. Further, behavioral experiments were conducted while the child was exposed to the VR stimuli (e.g., What do you think the dog might do if the dog was off the leash?) to assist in challenging the child dysfunctional phobic beliefs. The psychoeducation and nature of the behavioral experiments was largely standardized across participants, but flexibly delivered depending on the child's progress and pace of exposures. At the end of every hour, ratings

were taken of the children's perception of how real the VR stimuli was.

In the last approximately 30 minutes of the session, the researcher invited parent/s into the session to discuss the child's progress, topics presented in psychoeducation, the importance of ongoing in-vivo exposure, modeling and verbalization of nonanxious behavior by the parent. Furthermore, the maintenance phase of therapy was discussed with parents and the importance of weekly homework exposure practice. After the completion of the VR OST, the children were asked to complete the postassessment in vivo BAT.

Posttreatment, Maintenance Phase and Follow-up (1 Week and 1 Month)

At postassessment (1 week later), child and parent ratings were obtained over the telephone and the ADIS-P interview was conducted by a trained, independent assessor (postgraduate level clinical psychology student). During the maintenance phase, the therapist called the parent to discuss the child's progress, plan for further exposure tasks, and assist in any challenges the family were facing with ongoing exposures. The therapist worked with the child and parent to collaboratively develop homework exposure exercise. The maintenance calls were brief (limited to 15 to 30 minutes) and also involved in ongoing collection of baseline data. At 1-month follow-up, the ADIS-P was conducted by an independent assessor over the phone, and the SCAS – P/C, FSSC-R-C, parent-rated target behaviour, and BAT were completed at the clinic by the treating therapist.

Results

OVERVIEW OF ANALYSES

Single-case data were examined via visual inspection of CSR and Parent Target Behavior ratings across baseline, treatment, and follow-up periods, in line with guidelines for reporting single-case data (i.e., SCRIBE Statement; Tate et al., 2016). Stability over the baseline phase was examined by way of nonparametric Wilcoxon rank-sum tests (pre-assessment to final baseline point). Given the small sample size, nonparametric Friedman tests were used, followed by post hoc Wilcoxon rank-sum tests, to examine symptoms at baseline, compared to symptoms at post and at 1-month follow-up (CSR, parent-rated target symptoms and behavioral approach). Wilcoxon tests were used when the Friedman test was significant to determine incremental changes across each time point. Wilcoxon tests were also conducted on self-reported outcome measures (SCAS-C/P and FSSR-C) from pre-assessment to 1-month follow-up. Hedges's *g* was reported as an estimate of effect size, including 95%

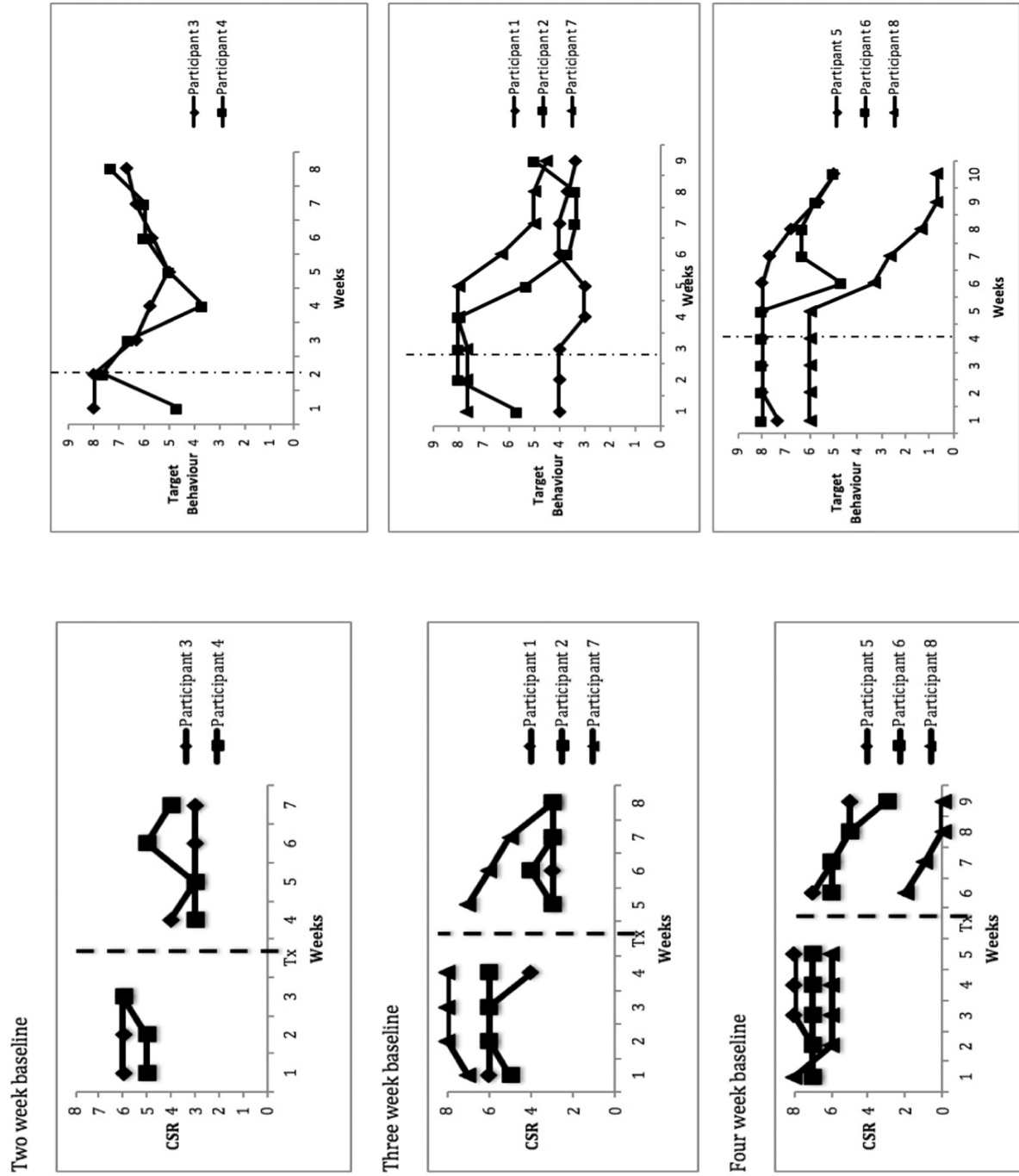


FIGURE 2 Single case Clinician Severity Rating (CSR) and Target Behavior Ratings across time for participants assigned to the 2-week, 3-week and 4-week baseline condition (including initial pre-baseline assessment for CSR).

confidence intervals, across time from pre- to posttreatment, pre- to 1-month follow-up, and posttreatment to 1-month follow-up. The formula for calculating point estimates for Hedges's g effect sizes was derived from Lakens (2013) for paired samples t -tests, using the average of the standard deviations method (see Uanboro, 2017). A Reliable Change Index (RCI; Jacobson & Truax, 1991) was calculated to determine whether the magnitude of change in children's SP severity (CSR) was statistically reliable. An RCI cut-off of 1.96 standard deviation units was used to meet criteria for reliable improvement. Finally, children were considered "recovered" if following treatment their ADIS CSR was below 4 and they met criterion for reliable change on the SP CSR.

SINGLE-CASE BASELINE DATA

There were no significant changes in symptoms over the baseline period for children (including 2-week, 3-week or 4-week baseline monitoring) on CSR ($Z = -.108$, $p = .41$), or on parent-rated target symptoms ($Z = -.108$, $p = .19$). Visual inspection of the single-case data demonstrates stability across the baseline, with decline following the VR EXP on both CSR and parent-rated target symptoms. Figure 2 illustrates the multiple baseline data for CSR and parent-rated target symptoms respectively, for participants assigned to the 2-week, 3-week, and 4-week baseline monitoring conditions.

WITHIN SUBJECTS EFFECT OF TIME

Across outcome measures, there were significant within-subjects effects for time, with reductions from pretreatment to posttreatment and 1-month follow-up on CSR ratings, $\chi^2(2) = 12.28$, $p = .001$; parent-rated target symptoms $\chi^2(2) = 8.96$, $p = .007$; and BAT, $\chi^2(2) = 12.28$, $p = .003$. A significant reduction in CSR was found from pre- to posttreatment ($Z = -2.26$, $p = .02$), pretreatment to 1 month ($Z = -2.53$, $p = .004$), and from posttreatment to 1-month follow-up ($Z = -1.89$, $p = .047$). For parent-rated target symptoms, there was a significant reduction from pre- to posttreatment ($Z = -2.19$, $p = .02$), pretreatment to 1 month ($Z = -2.52$, $p = .004$); however, there was no significant change from posttreatment to 1-month follow-up ($p = .29$). For behavioral approach as measured by the BAT, there was a significant reduction from pre- to posttreatment ($Z = -2.21$, $p = .02$), pretreatment to 1 month ($Z = -2.38$, $p = .008$); however, there was no significant change from posttreatment to 1-month follow-up ($p = .50$). For self-report measures, there were no significant effects of time from pretreatment to

1-month follow-up for SCAS-C ($p = .34$), SCAS-P ($p = .31$) or the FSSC-R ($p = .34$), indicating no overall reduction in parent- or child-reported anxiety or child-reported fear. Table 3 presents the means, standard deviations, 95% confidence intervals, and outcomes for the within-subjects repeated measures effects of time using Friedman's tests and Wilcoxon tests. Estimates of effect sizes across each time point on all measures are included with Hedges's g and 95% confidence intervals given the small sample size.

Reliable Change and Diagnostic Outcomes

A reliable change index (RCI) (Jacobson & Truax, 1991) was calculated for CSRs (RCI = 1.91) using the test-retest reliability coefficient obtained from Silverman, Saavedra, and Pina (2001). The results indicated that 63% ($n = 5$) of participants met the criterion for reliable change in CSRs from pretreatment to posttreatment, with 88% ($n = 7$) of participants at 1-month follow-up. In order for a participant to be considered "recovered" at study end point (1-month follow-up), they were required to have an ADIS CSR below clinical cut-off for their SP of dogs (ADIS, CSR < 4) in addition to meeting the criterion for RCI. Results indicated at 1-month follow-up, 75% ($n = 6$ of 8) of the sample were considered recovered (RCI and CSR < 4). In relation to overall diagnostic comorbidity, there was no change in the total number of children's diagnoses (excluding the treated SP) from pretreatment ($M = 2.00$, $SD = 2.92$) to 1-month follow-up ($M = 1.87$, $SD = 2.99$), $Z = -1.00$, $p = 0.50$.

WITHIN SESSION RATINGS

Within Session Reality/Immersion Rating

Level of reality/immersion of the VR stimuli ranged from zero to four ($M = 1.62$, $SD = 1.30$). Mean scores indicated that the level of reality/immersion of the VR was "a little bit like real life" across each hour, with no significant differences across hourly ratings (p 's > 0.05). Correlations were examined between level of reality/immersion and pre- to post-change scores on CSR, target symptoms and behavioral approach, as well as on change scores from pre- to 1-month follow-up across measures; however, no significant correlations were observed (change scores: CSR r 's = .21 – .51; target symptoms r 's = .07 – .45; behavioral approach r 's = -.57 – .00).

Discussion

The current study represents the first controlled case series to examine the preliminary efficacy of a VR OST for children with SP of dogs. The aim of this pilot study was to determine the feasibility, acceptability, and preliminary effectiveness of this novel approach. This study makes an important

Table 3
Means, Standard Deviations, Confidence Intervals and Effects of Time for Treatment Outcome Measures

Measure	Pre-M, SD	End Baseline M, SD	Post-Treat M, SD	Hedges <i>g</i> pre- to post, (95% CI)	One-month M, SD	Hedges <i>g</i> pre- to 1 month (95% CI)	Hedges <i>g</i> post- to 1 month (95% CI)	Freidman χ^2 / Wilcoxon Signed Ranks (p-value)
CSR	6.37 (1.06)	6.55 (1.33)	4.37 (1.99)	1.12 (0.17, 2.28)	3.00 (1.41)	2.40 (1.12, 4.24)	0.75 (-0.01, 1.56)	12.28 (.001) **
Target Symptoms	7.08 (1.41)	7.00 (1.91)	5.08 (1.69)	1.14 (0.27, 2.25)	4.68 (2.04)	1.29 (0.22, 2.37)	0.20 (-0.78, 1.18)	8.96 (.007) *
Behavioral Approach	3.50 (3.77)	-	8.75 (3.53)	-1.27 (-2.48, -0.33)	9.62 (1.06)	-1.96 (-3.6, -0.74)	-0.30 (-1.10, 0.46)	12.28 (.003) **
SCAS-C	35.00 (20.61)	-	-	-	32.87 (11.86)	0.11 (-0.48, 0.72)	-	-0.49 (.336)
SCAS-P	22.00 (14.78)	-	-	-	20.62 (17.45)	0.07 (-0.20, 0.36)	-	-0.56 (.309)
FFSC-R	115.37 (27.33)	-	-	-	112.50 (26.74)	0.09 (-0.42, -0.63)	-	-0.49 (.336)

Note: CSR = ADIS-P Clinician Severity Rating; SCAS-C/P = Spence Children's Anxiety Scale – Child / Parent report; FFSC-R = Fear Survey Schedule for Children – Revised; CI = 95% confidence intervals; M = Mean; SD = Standard Deviation; ** $p < 0.005$; * $p < 0.01$

contribution in regards to the provision of alternative modalities of treatments available for SP, which may improve uptake by clinicians in delivering evidence-based exposure therapy, as well as increased access to care for patients. It was hypothesized that following a VR OST, phobic symptoms, severity, and approach behaviors would improve and remain improved over a 1-month follow-up. Prior to commencing treatment, participants were mostly stable on clinician-rated diagnostic severity and parent-rated symptom severity across baselines. Analyses indicated a significant decrease in symptom severity from pretreatment to both posttreatment and 1-month follow-up (CSR and target symptoms), with CSR continuing to significantly decline across the maintenance phase following VR OST. Notably, VR-OST was also associated with clinically significant outcomes. At 1-month follow-up, 88% of the sample was reliably improved on CSR, and 75% of participants were considered “recovered” of their SP. Moreover, significant behavioral changes were also observed. At pretreatment no child was able to make physical contact with an unfamiliar dog under experimental conditions (i.e., BAT), yet at posttreatment and at the 1-month follow-up, 88% (7 out of 8) of children were able to fully approach and pat the dog on the head for longer than 20 seconds. These observed clinical indicators of treatment response are largely consistent with findings of previous studies examining SP in multiple-baselines studies of modified approaches to the OST (e.g., play modified OST, Farrell, Kershaw, and Ollendick, 2018; Farrell, Waters, et al., 2018; OST for BII, Oar et al., 2015) and provide support for the study aims and hypotheses.

However, child- and parent-reported anxiety (SCAS-C/P) and child-rated fear (FFSC-R) did not significantly reduce over the course of the study, despite study hypotheses. This finding is inconsistent with previous studies that have found significant improvements on the same measures following in-vivo OST for SP (Farrell, Kershaw, and Ollendick, 2018; Farrell, Waters, et al., 2018; Oar et al., 2015; Ollendick et al., 2009; Öst et al., 2001). The findings are, however, not surprising, given the assessments of general anxiety and broad fears, and the majority of this sample presented with mostly SPs (only 2 children experienced another anxiety disorder other than SP). Assessing treatment effects over a longer period of follow-up, with a larger clinical sample, would be necessary to determine whether these secondary benefits are obtained over a longer period of time.

In addition to treatment outcomes, this study also examined the level of reality/immersion in the VR

OST, as this may be associated with favorable outcomes in VR studies (Krijn et al., 2004; Robillard, Bouchard, Fournier, & Renaud, 2003). However, this study showed no correlation between the level of reality and treatment outcomes, which is consistent with Krijn et al. (2004) and Moldovan and David (2014). Given the relative infancy of VR therapies, there is a clear need for larger trials aimed at examining mediators of response, including whether “immersion” is a critical component.

STRENGTHS AND LIMITATIONS

Strengths of this study included the study design, which provided within-subjects experimental control using gold-standard assessment procedures. Furthermore, the treatment was guided by standardized, evidence-based OST (see Davis et al., 2012, 2019). Additionally, a 1-month follow-up provided insight into the potential durability of outcomes at least on a short-term basis. Further, the use of 360-degree footage is a more recent development of VR and provides an immersive experience designed to be more realistic than previous technology.

There are several limitations to the current study, including the small clinical sample, which limits the study power, the generalizability of findings, and does not allow for comparability of VR OST to other treatments (i.e., in-vivo OST). Furthermore, the treatment delivered involved (minimal) therapist-assisted homework exposure therapy following VR OST; thus, the relative contribution of the maintenance phase cannot be determined by this design. The inclusion of homework exposure-practice following VR OST remains consistent with the standard, in-vivo OST approach for children, in that OST is considered a “kick-start” to treatment, and ongoing exposure practice is considered essential for children to consolidate gains, make further gains, and maintain progress. Data from the BAT provide immediate evidence for the efficacy of the VR OST component, relative to the combined VR OST and in-vivo exposure homework maintenance phase. Future studies that more comprehensively assess outcomes immediately following VR OST, and again following the maintenance phases would provide further information on the relative efficacy of these two complimentary therapeutic phases. Finally, large RCTs of VR OST are needed to determine the relative efficacy of VR OST over in-vivo OST and alternate conditions.

IMPLICATIONS AND CONCLUDING REMARKS

In-vivo exposure therapy is underimplemented in routine clinical practice for a number of reasons,

such as the burden associated with sourcing and managing multiple phobia stimuli within a clinic setting, therapist anxiety/discomfort, lack of training/supervision, cost and time constraints. VR exposure therapy may provide a more feasible and acceptable modality for community clinicians, as well as for patients. While there is cumulative evidence for the efficacy of VR exposure among adults with anxiety and phobias, there is comparatively limited research with children or studies in community practice. This study provides initial support for VR OST as an effective approach in reducing children’s SP of dogs. Our results show promise for practitioners to use VR exposure treatment for treating childhood SP, and provide a cost-effective, comfortable, practical and feasible treatment. VR exposure may also be a useful treatment adjunct, prior to in-vivo exposure for patients reluctant to engage in in-vivo exposure, or following in-vivo exposure to facilitate homework compliance.

Future research may consider approaches to enhancing outcomes, via exploring the use of different VR contexts, more interactive VR, and across different SP subtypes. Although the treatment showed positive improvement, the precise mechanisms of how VR exposure may exert its effects are also not yet known. Future research aimed at exploring mechanisms of VR exposure therapy efficacy may assist in developing more potent VR approaches. Furthermore, trials of VR OST in community settings are also needed to determine feasibility, acceptability, and effectiveness when delivered by community clinicians in routine practice (e.g., Farrell, Barrett, & Claassens, 2005). Indeed, there is a great need for future studies to examine predictors, moderators, and mediators of change associated with VR exposure. Findings of this nature would inform greater precision in the development and delivery of VR exposure therapy, and ultimately result in enhanced patient outcomes.

There are a variety of phobias where exposure can be expensive, impractical, and/or uncomfortable to perform such as for flight phobia or blood injection phobia. If VR OST can be made more readily accessible and affordable to community clinicians, it may assist in more timely and effective treatment for children with a SP, and also lead to possible prevention of future mental health illnesses. In conclusion, this study provides promising outcomes for VR OST for children with a SP of dogs and supports a larger scale efficacy trial relative to in vivo OST and an effectiveness trial with community clinicians.

Conflict of Interest Statement

The authors declare that there are no conflicts of interest.

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